Lesson 18: Absolute Convergence, Conditional Convergence

Assessment Criteria & Rubrics

- **Definition & Initial Setup:**
  - **Excellent:** Correctly states that absolute convergence requires $\sum_{n=1}^{\infty} |a_n|$ to converge.
  - **Passing:** Attempts to take the absolute value of the terms before applying tests.
- **Test Selection & Application:**
  - **Ratio Test ($\lim_{n\to\infty} |\frac{a_{n+1}}{a_n}| = L$):** Best for factorials and exponentials.
    - **Absolute Convergence:** $L < 1$.
    - **Divergence:** $L > 1$.
    - **Inconclusive:** $L = 1$.
  - **Root Test ($\lim_{n\to\infty} \sqrt[n]{|a_n|} = L$):** Best for terms raised to the n-th power.
    - **Absolute Convergence:** $L < 1$.
  - **Comparison Tests:**
    - **Direct Comparison:** Shows $|a_n| \leq b_n$ where $\sum b_n$ is a known convergent series.
    - **Limit Comparison:** Evaluates $\lim_{n\to\infty} \frac{|a_n|}{b_n}$ to show convergence.

- **Conclusion Formulation:**
    - **Absolute Convergence:** If $\sum|a_n|$ converges, definitively state $\sum a_n$ is absolutely convergent (and thus convergent).
    - **Conditional Convergence:** If $\sum|a_n|$ diverges but $\sum a_n$ converges (e.g., via Alternating Series Test), state the series is conditionally convergent.
    - **Divergence:** If $\lim_{n\to\infty} a_n \neq 0$, the series diverges ($n$-th term test).